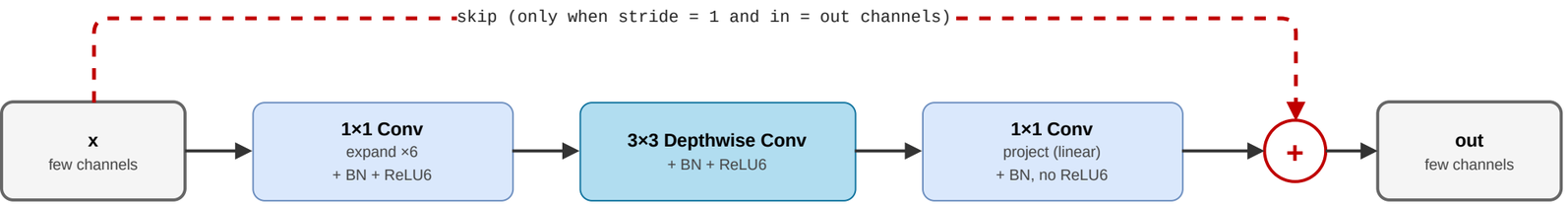
MobileNetV2 — lightweight · input 224×224, output n_classes logits

Input / Output | Conv (+ BN, activation) | Group of blocks | Depthwise Conv | Pool + classifier | **red dashed line = skip connection**

1. Overall architecture

2. Inside one Inverted Residual block (narrow → wide → narrow)



Key points for viva

- Depthwise separable convolution (3×3 depthwise + 1×1 pointwise) gives far fewer parameters and FLOPs than a normal conv.
- Inverted residual: the skip connects the thin (bottleneck) ends, and the middle is expanded ×6; the last 1×1 is linear so information is not lost.
- ≈ 2.3 M parameters with n_classes = 43.